Link

USER MANUAL

Wireless remote for your hi-fi



A wireless remote for your hi-fi.

Phone, tablet and computer.

Firmware 2.3.80 · October 2026

NOTE

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01

What Link does

Link is a small box that plugs into your preamplifier's RS-232 control port and puts it on your home network. From then on, any phone, tablet or computer in the house is a remote control.

```
Phone · Tablet · PC  ──Wi-Fi──▶  Link  ──RS-232──▶  Preamplifier
```

- **Volume, mute, inputs, tone, outputs** — everything the front panel does.
- **Always in sync.** Link mirrors the preamplifier. Turn the real knob and every screen follows.
- **A volume ceiling you set**, enforced inside Link — not just in the app.
- **Nothing leaves your home.** No account, no cloud, no subscription. The only thing Link ever fetches from the internet is its own software updates.
- **More than the preamp:** subwoofers, room correction, a disc player, a music streamer's transport controls.

02

What you need

- Your preamplifier, with an **RS-232 control port** on the back — a 3.5 mm jack on current models, a 9-pin DB9 on a few older ones (see Compatible equipment).
- A **Wi-Fi network** (2.4 GHz).
- A phone with the free **Link app** (App Store / Google Play) for the one-minute setup. After that, any phone, tablet or computer with a modern browser is a remote.
- A **USB-C charger**, 5 V, 0.5 A or more. Any phone charger will do.

In the box: the Link unit, one control cable and one USB-C cable. The control cable is the one chosen at order: 3.5 mm to 3.5 mm for most models, or 3.5 mm-to-DB9 for **equipment with a DB9 port**. A second or third 3.5 mm device needs its own stereo (TRS) cable — any ordinary male-to-male "aux" cable works. A DB9 cable is wired tip to pin 2, ring to pin 3, sleeve to pin 5 — the same as the manufacturer's own RS232 data cable — if you need to make another.

03

Connect it



Rear panel

Left to right on the back: the **control ports**, the **USB-C** power socket, the **status light** and the **button**.

1. **Control cable.** Run the supplied 3.5 mm cable from **any PORT** on Link to the **RS-232** jack on the back of your preamplifier. It sits in the row of small jacks, usually next to IR IN. Any port works and the order does not matter — Link finds out for itself what is plugged in where.
2. **More devices.** A second preamplifier, a B&W DB1 subwoofer, a MEN220 room-correction processor or a McIntosh MVP disc player each get their own cable into any free port.
3. **Power.** Plug in USB-C last. Link has no power switch — it is designed to stay on. It draws less than a phone on charge.

NOTE

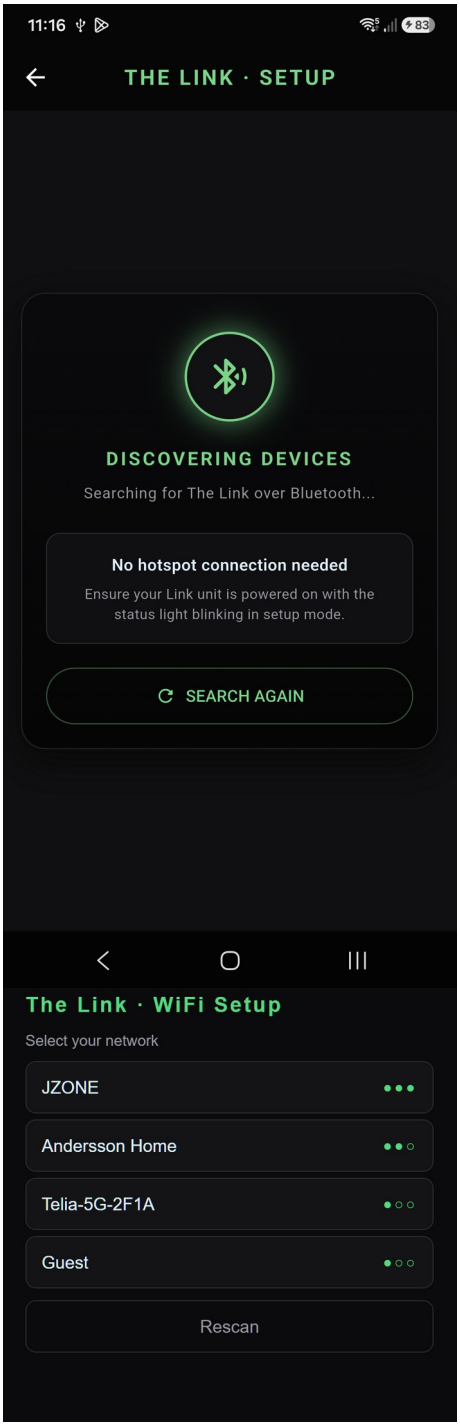
Leave the preamplifier's RS-232 baud rate at its factory setting (115200). If someone has changed it in the setup menu, Link will still find it — it tries every rate — but the factory setting is fastest.

04

First-time setup

When Link powers up for the first time its light **blinks blue, twice, pause, repeat**. That means it is waiting to be set up. Your phone never has to leave your home Wi-Fi.

With the Link app (recommended)



1. Install **Link** from the App Store or Google Play and open it. Allow Bluetooth when asked — it is used only for this step.
2. The app looks for a Link on your network, finds none, and starts a **Bluetooth search** by itself. Stand within a few metres of the unit.
3. It finds the unit and shows the **Wi-Fi networks the unit can see**.
4. Pick your home network and type its password. Optionally **name this Link** (do this if you own more than one — the name becomes its web address, e.g. `Studio` → `http://studio.local`).
5. Tap **Connect**. Link joins your network, the light goes **solid green**, and the app lands on the remote.

If the password was wrong the app tells you after about a minute and lets you retype it.

Without the app (fallback)

Use this if you cannot use Bluetooth, or if the app says **"Can't find it?"** — it offers the same path as **SET UP OVER WI-FI**.

1. On your phone, tablet or computer, join the Wi-Fi network **`Link`** — password **`linksetup`**. Accept any "no internet" warning.
2. A setup page usually opens by itself. If not, open a browser and go to **`http://192.168.4.1`**.
3. Tap your home network, type its password, optionally name the unit, and tap **Connect**. Link restarts, joins your network and the light goes solid.
4. Put your device back on your home Wi-Fi.

05

Opening the remote

The remote is a web page that Link itself serves. Nothing to download — the Link app simply hosts that same page, and a browser reaches it directly at **`http://link.local`**, or `http://<the-name-you-chose>.local` if you named the unit during setup.

Add it to your home screen and it opens like an app, full screen:

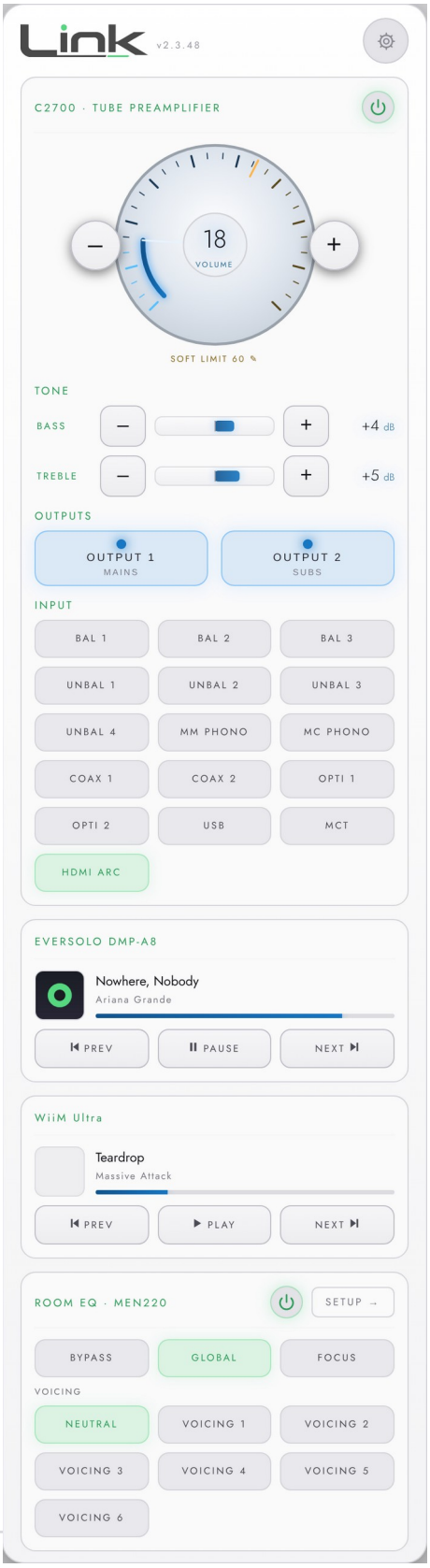
iPhone / iPad	Safari → **Share** → **Add to Home Screen**
Android	Chrome → menu → **Install app** / **Add to Home screen**
Windows / Mac	Chrome or Edge → address-bar menu → **Install**

Every phone, tablet and computer in the house can be open at once; they all show the same thing, because Link — not the app — owns the state.

06

The main screen

Everything is on one screen. Each piece of equipment gets its own card.



The preamplifier card

Power — the ⏻ button in the card header. When the preamplifier is off, the controls below dim; the header stays lit. Tube preamplifiers take around 15 seconds to warm up.

Volume — the big dial.

- Tap **+** and **–** either side of it. **Hold** either one to run.
- On a computer you can also **scroll the wheel** over the dial, or use the **arrow keys**.
- **Tap the number in the middle to mute.** One tap, always — and one tap back. Mute never has a confirmation step.

There is deliberately **no drag-to-set and no tap-on-dial**. Volume only ever moves one step at a time, so a slip of the thumb can never jump it.

Soft limit — the line under the dial. See Volume safety.

Tone — bass and treble, ±12 dB. Tap the word **TONE** to switch the tone controls in and out of circuit. Your preamplifier stores tone settings *per input*, so they follow the source. Models without tone controls hide the section.

Surround — appears only on A/V receivers and processors (AUTO / THROUGH / MOVIE / MUSIC / GAME).

Outputs — OUTPUT 1 and OUTPUT 2, with a lamp each. Rename them to whatever they actually drive ("MAINS", "SUBS", "HEADPHONES") in Panel Setup.

Inputs — a button per input. Hide the ones you never use and rename the rest, so the panel shows *your* system and nothing else.

The streamer cards

If an Eversolo, a WiiM, a Bluesound (or another BluOS player from NAD, DALI or PSB), a Denon or Marantz HEOS speaker, an OpenHome player such as a Linn, Lumin or Auralic, a Cambridge Audio network player, or another

UPnP/DLNA music player such as a HiFi Rose is on your network, Link finds it and shows what's playing: cover art, title, artist, and a progress bar you can **tap to seek**. **PREV · PLAY · NEXT** control the transport.

Eversolo, WiiM and BluOS players are fully supported. On internet radio a BluOS player may not allow skipping back, and seeking works only where the player allows it. A HEOS speaker shows the track and takes play, pause and skip, but has no progress bar. OpenHome and Cambridge Audio players show the track and take play, pause and skip; skipping and seeking follow what the player's current source allows. Other UPnP/DLNA players work for transport control; their track information may be limited.

Link shows **every streamer it finds**, each with its own card — for example a WiiM in the living room and a network player in the study. To hide a card you don't want, switch off **SHOW ON PANEL** on that streamer's page; it hides that card alone. **REMOVE** on the page takes the player away altogether. A television is never taken for a streamer, even though most smart TVs can play music over the network.

Link deliberately does **not** control the streamer's own volume — the preamplifier owns volume, full stop.

The disc player card

Shown when a McIntosh **MVP861**, **MVP871** or **MVP901** is connected to one of Link's ports: what the player is doing (with an audio CD, the current **TRACK**), **PREV · STOP · PLAY/PAUSE · NEXT**, and power. Link finds the player by itself, like everything else on a port.

The room-correction card

Shown when a McIntosh MEN220 is connected. See section 9.

Several preamplifiers

Link can drive up to four preamp-class devices at once, one per port or over the network. When it detects a second one, **tabs appear at the top** of the screen. Each has its own volume, its own limit, its own input names — and its own safety apparatus. Two phones can sit on different tabs at the same time.

Themes

Six colour schemes — DARK, LIGHT, GRAPHITE, SILVER, CLASSIC and STUDIO — under **Panel Setup → SYSTEM → THEME**. The choice is per phone or browser, so everyone in the house can have their own. The pictures in this manual use LIGHT.

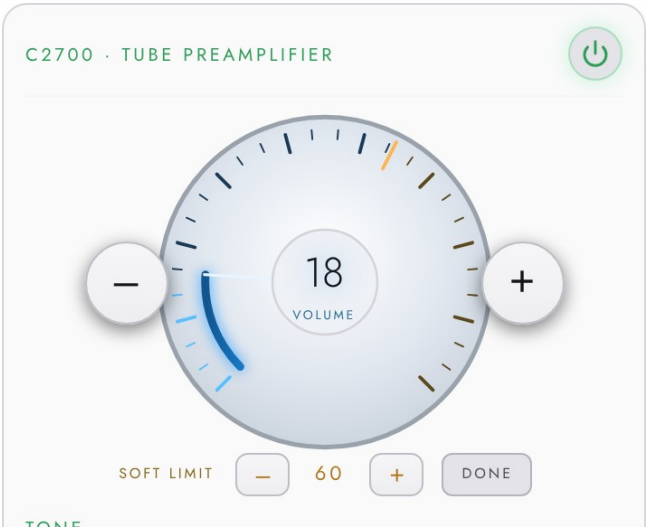
07

Volume safety

This is the part of Link that matters most, so it is worth a minute.

Link never sends an absolute volume. It only ever says "one step up" or "one step down", and it waits for the preamplifier to confirm each one. There is no code path in the product that can command a volume number — so no glitch, no stale screen and no lost connection can produce a jump.

The soft limit is a ceiling you set. Under the dial you'll see `SOFT LIMIT 60 ✎`. Tap it, adjust with **–** / **+**, tap **DONE**. Volume-up requests are simply dropped at the ceiling. This is enforced inside Link, not in the app, so it holds for every phone in the house.



Editing the soft limit

To go louder on purpose, wait until the volume reaches the limit — an **UNLOCK >60** button then appears under the dial. Tap it and the line changes to `HIGH ZONE UNLOCKED · relocks in 30 s`. After 30 seconds the ceiling comes back by itself. It is a deliberate, temporary act — never a setting you can forget you left on.

Lowering the limit below the current volume walks the volume down to meet it, one confirmed step at a time.

Other protections, always on:

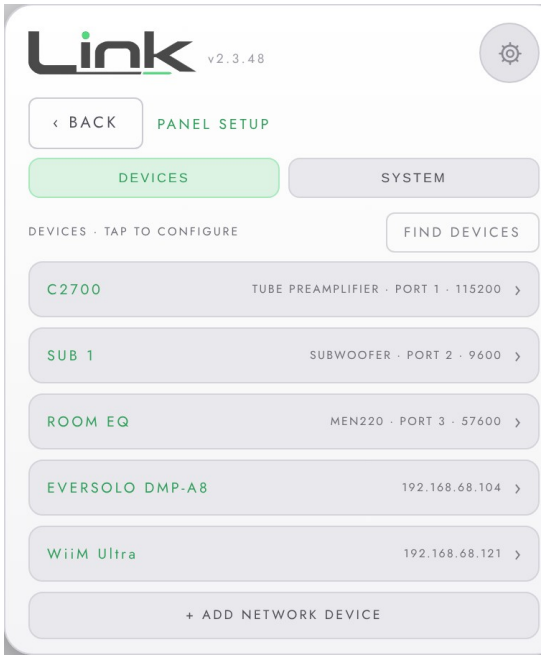
- A minimum interval between volume steps, so a stuck button or a flaky network can't sprint.
- Hold-to-repeat is the *app* sending single steps. Lose the connection and the volume stops moving that instant.
- Link never assumes. It shows what the preamplifier has confirmed.

> **NOTE**
> **Set the limit before you need it.** Pick a number that is loud but safe for your speakers, and leave it there.

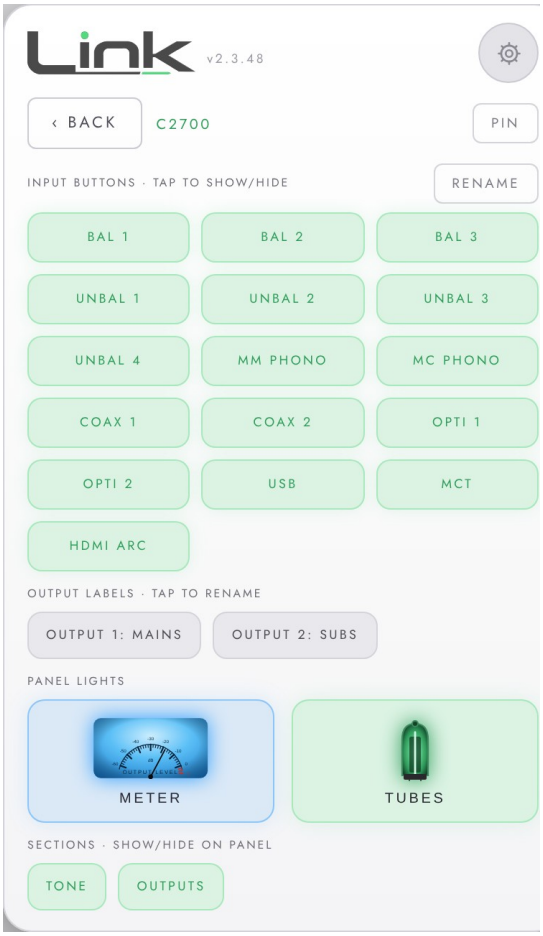
08

Panel Setup

Tap the ☰ **menu** in the top right, then **SETTINGS**. Two tabs: **DEVICES** and **SYSTEM**. The menu also has FIND DEVICES, THEME, CHECK FOR UPDATE, REARRANGE CARDS and ABOUT.



Panel Setup — devices



A preamplifier's own settings page

DEVICES

A list of what Link has actually found — each row shows which port it's on and at what speed. **Tap a row** to open that device's own settings page. Ports still searching appear dimmed at the bottom as "PORT n — scanning…"; that list *is* the diagnostic.

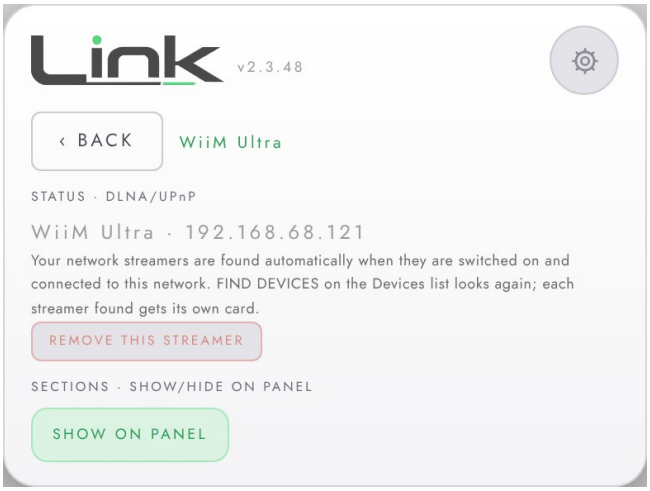
FIND DEVICES in the header makes Link look again — on its empty ports, on the network for streamers, and on the network for supported preamplifiers. It never removes a wired device it has already found; a device that has stopped answering stays in the list marked as lost until it answers again or Link restarts. It is the only "look again" button; everything Link finds appears in this list by itself. Each streamer gets its own row.

On a preamplifier's page:

- **INPUT BUTTONS** — tap to show/hide. **RENAME** to relabel them.
- **OUTPUT LABELS** — tap to rename.
- **PANEL LIGHTS** — METER and TUBES, the preamplifier's own front-panel lighting (models without them don't show the buttons).
- **PREAMP SETTINGS** — the preamplifier's own trim menu: balance, input trim, mono, display brightness, headphone HXD, and the MC and MM phono loads (adjustable while a phono input is selected). Models that have them also show phono gain, processor loop, output 1 trim and dual mono, and output 1's mode and crossover for reference — those two are set on the preamplifier itself. Only what your model reports appears here. These are stored in the preamplifier, not in Link, and separately for each input: the heading names the input they belong to. Display brightness is the same for all inputs.
- **SECTIONS** — show/hide TONE and OUTPUTS on the main screen.

The rest of this page is stored **in Link**, not in your browser — so a new phone and a tablet show the same layout. It is remembered per model, so a device keeps its names and its limit even if you move it to another port.

On a streamer's page: its name and address, **SHOW ON PANEL** (whether this player's card appears on the main screen; the other players are not affected), **OPEN WITH** — choose the app the streamer's card opens, for example Qobuz instead of the streamer's own app. (The card's open button asks the first time you tap it, too.) In the Link app on Android you choose from the apps installed on your phone; on an iPhone from a list of music apps. The choice belongs to the phone, so each phone can open its own app. The button only ever opens an app, never a web page — and **REMOVE THIS STREAMER** — to drop a player you no longer use. A later FIND DEVICES finds it again if it is still on the network.



A streamer's settings page

+ ADD NETWORK DEVICE adds equipment from other brands — see section 10.

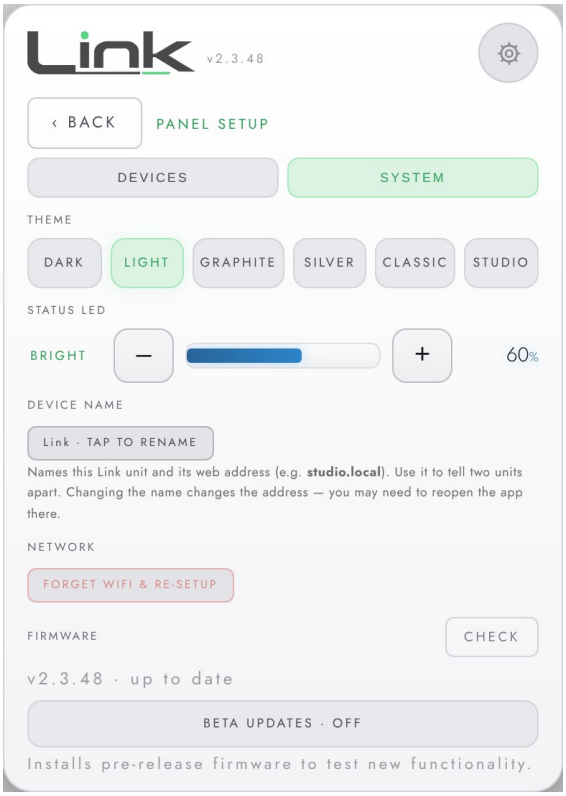
Pinning a setting to the main screen

Any setting on a device page — a subwoofer's preset, the room-correction voicing, the preamplifier's panel lights — can be put on the main screen, where it appears in a **PINNED** section at the bottom.

1. Open the device's page and tap **PIN** in its header.
2. Tap the setting you want. (While PIN is on, a tap pins the control instead of changing it.)
3. Tap **PIN** again to finish.

You can also **press and hold** a setting to pin or unpin it. A pinned setting is the very same control as on its settings page, with its current state shown. Like the rest of the layout, pins are stored in Link, so every phone sees them.

SYSTEM



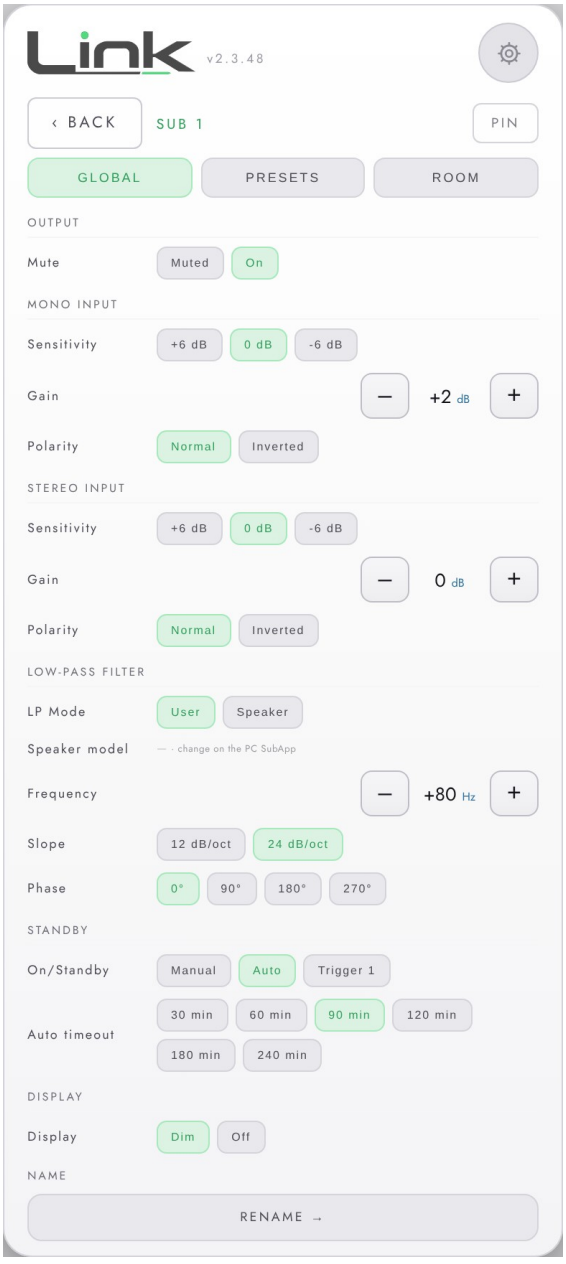
Panel Setup — SYSTEM

- **THEME** — the six colour schemes.
- **STATUS LED** — the brightness of the light on the box.
- **DEVICE NAME** — names this Link unit *and* its web address. Use it to tell two units apart. Changing it changes the address, so you may need to reopen the app at the new one.
- **NETWORK** — **FORGET WIFI & RE-SETUP** to start the setup over.
- **FIRMWARE** — the running version, and **CHECK** for updates. See section 12.
- **BETA UPDATES** — off by default. Switch it on only if you want to try pre-release software before everyone else.

09

Subwoofers and room correction

Subwoofers. Link supports a pair of **B&W DB1** subwoofers on their own control links — one cable each into any free port. Once found, they appear in **Panel Setup → DEVICES** with a near-complete replica of the manufacturer's own settings app: gain, sensitivity, polarity, low-pass filter, standby behaviour, presets, mute — plus a rename.



Subwoofer settings

Room measurement itself stays on a PC — it needs the manufacturer's measurement microphone hardware.

Room correction. A McIntosh **MEN220** room-correction processor appears as a **ROOM EQ** card on the main screen:



Room EQ card

- **Correction mode** — Bypass, Global, or one of the Focus positions.
- **Voicing** — Neutral plus six alternative voicings.
- Power, the focus position and rename live on its **SETUP** page.

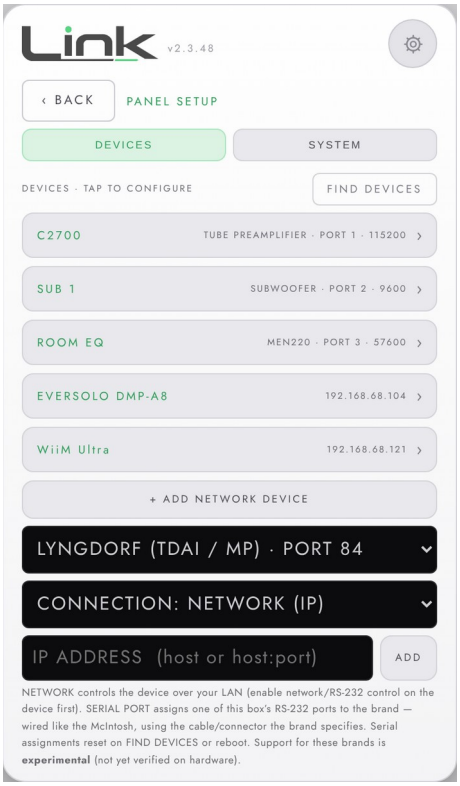
Focus positions are the ones measured with the processor's own guided setup and microphone; Link selects between them, it does not measure them.

10

Adding other brands

Link speaks more than one brand. Preamplifiers, integrated amplifiers and processors from **Lyngdorf, Rotel/Michi, Denon/Marantz, NAD, Yamaha, Onkyo/Integra, Audio Research, Arcam, Anthem, Mark Levinson, Primare, Meridian, Trinnov/JBL Synthesis, Krell** and **Classé** can be added — over the network, or wired to one of Link's own ports.

Panel Setup → DEVICES → + ADD NETWORK DEVICE



1. Pick the **brand**.
2. Pick the **connection**: NETWORK (IP), or one of Link's ports.
3. For a network device, type its **IP address** (add `:port` only if you have changed it from the brand default). Tap **ADD**. You rarely need this: Link looks for supported brands on your network by itself — once after start-up, at a moment nobody is using the remote, so you never feel it — and again whenever you press **FIND DEVICES**, and adds what it finds. The form is for a device that does not answer that search, for example one that ignores network "ping" or sits on another network.

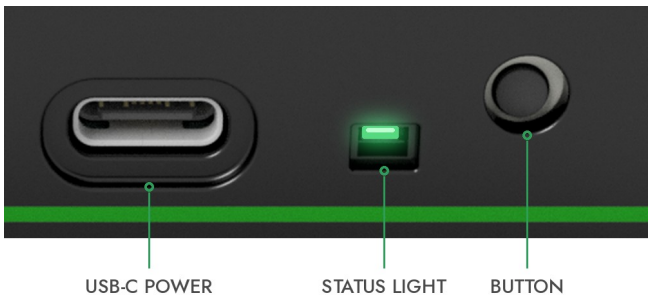
A receiver that is also a network music player — a Denon, Marantz, Yamaha, Onkyo, Integra, Arcam or Lyngdorf — is added as a preamplifier by itself as soon as Link finds it, alongside its streamer card. If you **REMOVE** a network device, Link remembers that and never adds it back by itself; adding it by hand undoes that.

Notes:

- Enable **network control** (sometimes called "IP control" or "standby network") on the device first, or it will stop answering when it sleeps.
- Volume on every brand is relative-only and limit-protected, exactly as described in section 7. The dial shows that brand's own scale — decibels where the device works in decibels.
- **Port assignments reset** on reboot; network devices are remembered.
- Support for these brands is **experimental** — written from each manufacturer's published control documentation, and not yet verified against a real unit. Try it at a low volume first.

11

The button and the light



Button and status light

The button and the light sit next to the USB-C socket.

The button

PRESS	DOES
Short press	**Sleep.** The whole unit switches off: light out, radio off. Press again to wake it.
Hold 5 seconds	**Factory reset.** The light flashes **red** while you hold; at 5 s it goes **dark** — that is your cue to let go.

> **NOTE**
> **While Link is asleep it is off the network, and the app cannot wake it.** Only the button can. It draws almost nothing asleep; if you would rather it were always reachable, simply never press the button.

A factory reset erases **everything**: Wi-Fi, device names, limits, layout — and reopens the `Link` setup mode. It never touches your preamplifier's own settings. Releasing before five seconds never resets: a short press is sleep, nothing more.

The status light

RHYTHM	COLOUR	MEANING
Solid	green	All good — on the network, talking to your equipment
Slow breathe	amber	On the network, waiting for the equipment to answer. Check the cable.
Fast breathe	amber	Can't join the saved Wi-Fi network
Two quick blinks, pause	blue	Setup mode — waiting to be configured
Fast flashing	red	You're holding the button. Let go now to cancel.
Off	—	Asleep, or no power

12

Updates

Link updates itself. It checks about once a day and installs a new version while the remote is idle. If you open the app while one is waiting, the FIRMWARE line in **Panel Setup → SYSTEM** shows **UPDATE** — tap it to install right away.

- Updates are **cryptographically signed** and verified inside Link before anything is installed. An unsigned or tampered update is refused.
- The update takes about 20 seconds. **Your music keeps playing** — the preamplifier holds its own state and Link re-syncs on the way back up. You simply can't change the volume from your phone while it happens. The page reloads itself when the new version is up.
- If a new version misbehaves, Link **puts the previous one back by itself**.
- **Beta updates.** Switch on **Panel Setup → SYSTEM → BETA UPDATES** to receive pre-release versions as well. A beta unit always runs the newer of the two; switching it off again keeps the version you have until the next regular release overtakes it.
- The app on your phone updates alongside it; you may see a *"A new version of the app is available — RELOAD"* bar. Tap RELOAD.

This is the only feature that talks to the internet. There is no telemetry, no account, and no cloud control.

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Keyboard shortcuts

On a computer, with the remote in focus:

KEY	DOES
↑ / ↓	Volume up / down (hold to repeat)
Scroll wheel over the dial	Volume
M	Mute
1 – 6	Select input 1 to 6
Space	Play / pause the streamer
N / P	Next / previous track

14

Troubleshooting

The app says NO LINK NEARBY. Check the light: it should be blinking blue. If it is solid green, the unit is already on a network — tap **ENTER AN ADDRESS** and type `link.local`. If it is dark, it is asleep: press the button once. If it breathes amber, it is trying to join a network it can no longer reach: hold the button until the light goes dark and set it up again.

`link.local` doesn't open. Some Android versions and older Windows setups can't resolve `.local` names at all. Open your router's device list, find `link` (or the name you gave it), and use its IP address directly — then bookmark it. If you named the unit, its address is the name in lowercase with spaces as hyphens: `Living Room` → `http://living-room.local`.

The app opens but says it isn't connected to the equipment. Check the control cable at both ends, and check the preamplifier is on. Then **Panel Setup → DEVICES → FIND DEVICES**. A device that is still searching shows as "PORT n — scanning…".

Volume won't go higher. You're at the soft limit. Tap **UNLOCK** under the dial, or raise the limit.

The volume moves in the app but nothing happens. The preamplifier isn't answering. Link only ever shows *confirmed* state, so if the number is moving, it is being confirmed — check the amplifier, not Link.

An input reports an error when selected. Inputs switched "Off" in your preamplifier's own setup menu refuse selection. Enable it there, or hide it in **Panel Setup**.

My streamer or network receiver isn't found. Make sure it is switched on and on the same network as Link, then **Panel Setup → DEVICES → FIND DEVICES**. Guest networks and separate network segments block the search; a supported receiver can still be added by hand with **+ ADD NETWORK DEVICE**.

Two units clash. Every unit answers to `link.local` until it is named. Give one a name in **Panel Setup → SYSTEM → DEVICE NAME** (or during setup). It then lives at its own address (`Living Room` → `http://living-room.local`), keeps its own settings, and leaves `link.local` free for the other one.

Starting completely over. Panel Setup → SYSTEM → FORGET WIFI & RE-SETUP, or hold the button on the box until the light goes dark.

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Compatible equipment

Link works with equipment that has a **modern 3.5 mm RS-232 control jack** with a selectable baud rate. It detects the model automatically — there is no picker — and adapts the screen to that model's inputs and features.

McIntosh preamplifiers	C2500, C2600, C2700, C2800, C47, C48, C49, C50, C52, C53, C55, C1100, C12000
McIntosh integrated amplifiers	MA5300, MA352, MA7200, MA8900, MA8950, MA9000, MA9500, MA12000
McIntosh processors / receivers	MHT300, MX100, MX200
McIntosh digital preamplifier	D1100
McIntosh disc players	MVP861, MVP871, MVP901 (transport and power)
DB9 port (choose the DB9 cable when ordering)	C48, C50, C2500, MEN220, MVP861, MVP871, MVP901, B&W DB1 — all others have the 3.5 mm jack
Other brands (experimental)	Lyngdorf, Rotel/Michi, Denon/Marantz, NAD, Yamaha, Onkyo/Integra, Audio Research, Arcam, Anthem, Mark Levinson, Primare, Meridian, Trinnov/JBL Synthesis, Krell, Classé

Also supported: B&W DB1 subwoofers · McIntosh MEN220 room correction · Eversolo, WiiM and Bluesound/BluOS streamers (full), Denon and Marantz HEOS speakers, OpenHome players (Linn, Lumin, Auralic) and Cambridge Audio network players (now playing and transport) and other UPnP/DLNA players (transport control).

Not supported: equipment with only a Data Port, an IR input or trigger outputs (for example the McIntosh C2200, C2300, MA7000 and the MVP831, MVP841 and MVP851 disc players). These are one-way connections — Link cannot read the state of the equipment through them, and Link never guesses. Link does not use infrared in any form.

If a receiver or processor has a **volume scale** setting, leave it on the **linear 0–99** scale.

16

Care, power and specifications

Ports	3 × 3.5 mm TRS, RS-232
Network	Wi-Fi 2.4 GHz, Bluetooth LE (setup only)
Power	USB-C, 5 V, 0.5 A minimum (cable included, charger not)
Cooling	Passive, silent
Enclosure	Compact desktop box

No user-serviceable parts and no fan. Keep it in open air — it is passively cooled and prefers not to sit inside a sealed cabinet or under another warm component. Wipe with a dry cloth. There is nothing inside to service.

Support

Questions: **info@playandtrack.com** — you write to the person who built it.

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